Testing Medical VLM Generalization on Rare Cases

MedGemma, BiomedCLIP, and MedSigLIP on Open-MELON

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Abstract

How well do medical VLMs generalize to rare histopathology cases? We evaluate whether caption-based MedGemma can classify melanoma vs. nevus on 913 Open-MELON histopathology images, and benchmark it against embedding-based classifiers (BiomedCLIP, MedSigLIP).

Methodology

Dataset. Open-MELON-VL-2.5K [1]: 913 samples (597 melanoma, 316 benign). Ground truth labelling by keyword matching.
Caption cleaning. Original captions contain non-visual metadata (staining, magnification, citations). Gemini rewrites each caption to retain only visual pathology and diagnosis, producing cleaned references for labeling and RAGAS evaluation.
Generative. MedGemma-1.5-4B-it [2] (Q4_K_M). Caption generation. Caption quality scored via RAGAS [5] (Gemini-judged faithfulness and relevance vs. cleaned references). 4 prompts × 3 decoding settings (deterministic, mild sampling at T=0.2, moderate sampling at T=0.5).
Embeddings. BiomedCLIP [3] (512-d) and MedSigLIP [4] (768-d) with LogReg, SVM, Random Forest.
Evaluation. Each trial samples equal numbers per class. We report balanced accuracy averaged over 10 trials. Failure theme analysis on shared hard cases.

Main Results

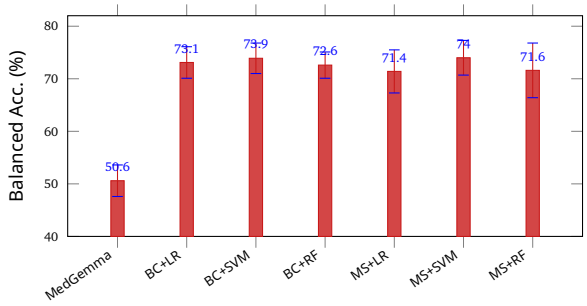

Figure 1: Balanced accuracy (10 trials). BC = BiomedCLIP, MS = MedSigLIP.

MedGemma bias. Binary-choice prompt: 99.4% melanoma recall but only 0.7% nevus recall. The model predicts melanoma for nearly every sample. Balanced accuracy: 50.6%.
Caption quality. The open prompt scores 0.15 faithfulness / 0.23 relevance; the binary-choice prompt improves to 0.44 / 0.57, but still indicates only partial alignment with reference pathology.

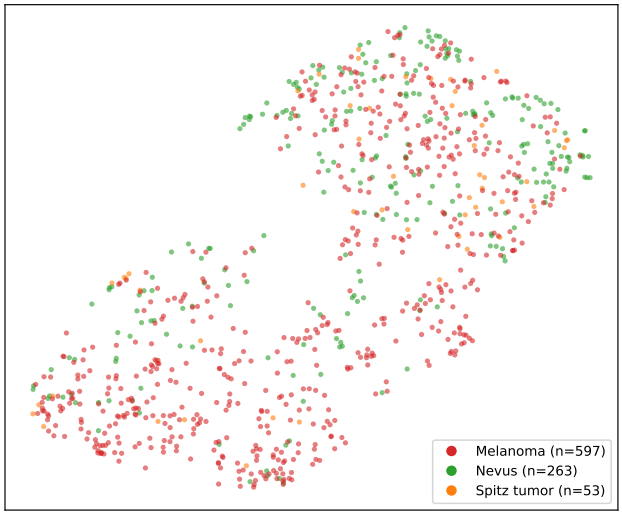

Figure 2: UMAP projection of MedSigLIP embeddings. Some separation between nevus and melanoma clusters is visible, but far from clean. Spitz tumor embeddings scatter across melanoma-dense regions, showing their features are indistinguishable from melanoma.

Discussion

- MedGemma fails to generalize: melanoma bias persists across all prompt configurations tested.
- We identified 35 cases where both BiomedCLIP and MedSigLIP SVMs fail. These concentrate in diagnostically challenging subtypes from both directions: benign lesions that mimic malignancy (Spitz tumors, cellular blue nevi) and melanomas that defy typical appearance (in-situ, mucosal, desmoplastic). These are cases that challenge even expert

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dermatopathologists, suggesting the models have reached a performance ceiling set by genuine morphological ambiguity.
- Labels extracted by Gemini from caption text, not from expert pathologist annotation, label noise is possible.

Conclusion

MedGemma does not generalize well. It overcalls melanoma on benign lesions regardless of prompt, prioritizing sensitivity over specificity.

References

[1] Han et al. (2024). Open-MELON-VL-2.5K.

[2] Google (2024). MedGemma.

[3] Zhang et al. (2023). BiomedCLIP. *arXiv:2303.00915*.

[4] Hamamci et al. (2024). MedSigLIP.

[5] Es et al. (2023). RAGAS. *arXiv:2309.15217*.